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PI Name:	Pandorf, Clay E Ph.D.		
Project Title:	Molecular Mechanisms that Impact Transcription of Myosin Genes in Human Skeletal Muscle in Response to Unloading: Role of Resistance Exercise		
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Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
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PI Email:	cpandorf@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-4289
Organization Name:	University of California, Irvine		
PI Address 1:	Physiology and Biophysics		
PI Address 2:	D360 Med Sci I		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-0004	**Congressional District:**	48
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Baldwin, Kenneth (MENTOR/University of California, Irvine)		
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	POSTDOCTORAL FELLOWSHIP Loading forces are critical to maintaining homeostasis in the muscle cell. Perturbations, such as microgravity, upset the normal gene expression of structural and signaling proteins. Unloading of skeletal muscle results in atrophy as well as deleterious alterations to the contractile properties and endurance capacities of muscle fibers. Exercise countermeasures can offset these alterations. Human skeletal muscle responds to exercise stimuli with alterations to myosin heavy chain (MHC) gene expression, and thus, contractile function. Original Project Aims: Aim #1: Does antisense RNA play a role in the transcriptional regulation of MHC genes in human skeletal muscle?		

Task Description:	Aim #2: Are histone modifications dynamically regulated at MHC genes during alterations to muscle loading state in human skeletal muscle? Aim #3: What are the functional performance and cellular-level adaptations to unloading and exercise training, and how does that relate to the corresponding molecular-level responses? Key Findings: Vastus lateralis biopsies were obtained before and after 5 wks of aerobic + resistance exercise (12 M, 9 F). The proportions of Type I and IIa MHC mRNA were both increased by 11%, while IIx was decreased by 38% after exercise training (all $p < .05$) in the vastus lateralis. Several novel discoveries were made regarding expression of long non-coding antisense RNA to MHC genes. As previously shown only in rats, lncRNA oriented antisense to the IIa MHC was detected and found to be down regulated with exercise (-55%; $p < .05$). A regulatory role is inferred from an inverse correlation between the proportion of IIa MHC mRNA and its antisense pair ($r = -0.66$; $p = .002$). Surprisingly we detected low levels of IIb mRNA and pre-mRNA, indicating that the IIb gene is not transcriptionally silent in human limb muscle, as thought. We also report the novel finding of as-lncRNA to the IIb gene. These data reveal that insights to MHC regulatory mechanisms previously shown in rats and involving as-lncRNA are conserved in humans. Two groups of healthy inactive human subjects participated in unilateral lower limb suspension (ULLS; male N=5; female N=4) for 10 days alone or with a combination of aerobic and resistance exercise training (ULLS+T; male N=5; female N=5). Soleus biopsies were obtained before and after ULLS. We examined gene expression of several large sarcomeric proteins that provide structural support to the myofilament (titin, nebulin and a-actin) and converge at the Z-disc. We also examined several signaling proteins that reside in this region (STARS, atrogin-1, and calcineurin via its modulator MCIP1). Nebulin, titin and a-actin RNA levels changed pre to post by -11%, +26% and +7% with ULLS, and by +25%*, +56%* and +16% with ULLS+T, respectively (*$p<0.05$ pre to post). STARS, atrogin1 and MCIP1 mRNA levels changed pre to post by -75%*, +77%* and -40%* with ULLS, and by -63%*, +30%* and -26%* with ULLS+T, respectively (*$p<0.05$ pre to post). While changes in transcription of the Z-disc associated structural proteins nebulin and titin following 10d unloading was not statistically significant, there was significant upregulation with unloading + training stimuli. The unfavorable unloading-induced signaling response was ameliorated when combined with training as suggested by differential mRNA levels of mechano-sensitive Z-disc factors that can promote atrophy (atrogin) and fiber-type shifts (calcineurin/MCIP1). These data suggest that exercise countermeasures to short-term unloading of the loading-sensitive slow soleus muscle can promote favorable transcriptional responses of proteins associated with the stress-sensitive Z-disc. In conclusion, these studies showed that a gravity-independent exercise device can induce alterations to key contractile and structural muscle genes, and can counteract some of the early alterations that occur with simulated microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Unloading of skeletal muscle results in atrophy as well as deleterious alterations to the contractile properties and endurance capacities of muscle fibers. This condition has significant impact on all humans with impaired muscle function due to disuse. These deleterious alterations are largely attributed to modification to the expression of the myosin heavy chain (MHC) genes. In particular non-fatigable, or slow, skeletal muscle motor units which provide for postural support and sustained contractile activity, are remodeled to fatigable, or fast, muscle, involving shifts from type I/IIa to IIx MHC in humans. Resistance exercise has been shown to effectively blunt these undesirable adaptations in both animals and humans. We have previously identified several novel mechanisms, such as antisense RNA and histone modifications, that impact expression of the MHC genes in unloaded rat muscle. Specifically, we showed that antisense RNA is implicated in transcriptional downregulation of IIa MHC gene during muscle inactivity. We also demonstrated that alterations to histone acetylation and methylation at specific MHC genes correspond with altered MHC gene transcription in response to muscle unloading. In the present studies, we were able to demonstrate that insights to MHC regulatory mechanisms previously shown in rats and involving long non-coding antisense RNA are conserved in humans. This may provide for development of novel therapeutics to curtail the slow to fast shift that occurs in disuse environments. We were also able to show that exercise countermeasures to short-term unloading of the loading-sensitive slow soleus muscle can promote favorable transcriptional responses of structural and signaling proteins in the muscle. Validation with humans of the mechanisms that regulate the genes of contractile and structural proteins in muscle should help to develop future strategies that most effectively ameliorate the effects of unloading on human skeletal muscle, as well as enhancing the understanding of these same processes that have significant impact on all humans with impaired muscle function.
Task Progress:	Twenty six subjects completed the unilateral lower limb suspension (ULLS) studies. These subjects participated in ULLS for either 10 or 5 days, with or without concurrent exercise training. Soleus and Vastus Lateralis muscle biopsies were obtained, and RNA and protein analyses were initiated. RNA analyses were also completed on muscle biopsies obtained last year with subjects that completed different types of exercise. We quantified the proportion of individual myosin heavy chain (MHC) mRNAs and protein in human muscle samples. The proportions of protein isoforms of MHC were determined from SDS-PAGE. mRNA proportions were determined with inclusion of a control DNA fragment that includes sequence from each of six human MHC genes which is co-amplified during PCR of cDNA from muscle samples. We measured mRNA, pre-mRNA, and antisense RNA from human muscle samples for the MHC genes. We measured mRNA levels of the genes of numerous other contractile and structural muscle proteins. Significant and important findings were made.
Bibliography Type:	Description: (Last Updated: 03/08/2012)
Articles in Peer-reviewed Journals	Pandorf CE, Jiang W, Qin AX, Bodell PW, Baldwin KM, Haddad F. "Regulation of an antisense RNA with the transition of neonatal to IIb myosin heavy chain during postnatal development and hypothyroidism in rat skeletal muscle." Am J Physiol Regul Integr Comp Physiol. 2012 Jan 18. [Epub ahead of print] PubMed PMID: 22262309 , Jan-2012
Awards	Pandorf C. "Beginning Investigator, April 2011." Apr-2011